Agriculture and Human Values (2026) 43:30
https://doi.org/10.1007/s10460-025-10805-1

# When the university meets the grassroots: participatory action research and just transitions in agriculture

**Ana Fochesatto**[1] · **Erin B. Lowe**[2] · **Adena R. Rissman**[3]

Received: 5 February 2025 / Accepted: 26 November 2025 / Published online: 28 January 2026
© The Author(s) 2025

## Abstract

Confronting the deep-rooted social inequities and environmental harm in animal agriculture research demands approaches that link Just Transitions (JT) with Participatory Action Research (PAR). We present a case study from the Midwest, USA, drawing on an ongoing participatory action research project, started in 2020, where we conducted semi-structured interviews with 128 participants and three participatory workshops with farmers, non-profit leaders, and community organizers. Adapting established PAR methods, we integrated community organizing tools such as one-on-one relational meetings and interactive workshops to better understand barriers to transition and how to foster meaningful change. These methodological interventions shifted who participated, how knowledge was produced, and whose voices shaped the terms of agricultural research and its implications for just transitions. They created deeper conversations about race, land, labor, and governance, which surfaced tensions while also creating the conditions for solidarity across difference. Grounded in this case study, we articulate a Just Transitions PAR Implementation Framework that highlights four interconnected elements: engagement and trust-building, collaboration and co-design, action and co-production of knowledge, and reflection and iteration. We argue that without intentional, justice-centered adaptations, participatory agricultural and transitions research risks reinforcing the very exclusions it aims to challenge. This paper contributes a conceptual intervention by positioning PAR as a valuable approach for advancing Just Transitions in agriculture and a practical methodological guide for researchers committed to transformative change from within and beyond the university.

**Keywords** Participatory action research · Just transitions · Racial justice · Grass-based agriculture · University · Food systems · Qualitative methods

**Abbreviations**

| | |
|---|---|
| PAR | Participatory Action Research |
| JT | Just Transitions |

## Introduction

Dominant agricultural systems are shaped by deep-rooted inequities of land dispossession, racialized labor regimes, and extractive systems that harm people and ecosystems (Guthman 2011; Horst and Marion 2019). These injustices are not incidental but foundational and reinforced by models like monocropping and Concentrated Animal Feeding Operations (CAFOs) that generate environmental harm while consolidating wealth and power (Horrigan et al. 2002; Hendrickson et al. 2020; Rissman et al. 2023). The call for Just Transitions in agriculture is urgent, yet efforts often remain technocratic, top-down, and disconnected from the

✉ Ana Fochesatto
fochesatto@wisc.edu

Erin B. Lowe
eblowe@wisc.edu

Adena R. Rissman
adena.rissman@wisc.edu

[1] Department of Community and Environmental Sociology, University of Wisconsin-Madison, 1450 Linden Dr., Madison, WI 53706, USA

[2] School of Human Ecology, University of Wisconsin-Madison, 1300 Linden Drive, Madison, WI 53706, USA

[3] Department of Forest and Wildlife Ecology, University of Wisconsin-Madison, 1630 Linden Drive, Madison, WI 53706, USA

lived experiences of farmers, workers, and communities on the frontlines of change (Just Transition Alliance n.d.; Stark et al. 2023; Wang and Lo 2021). Addressing systemic inequities requires confronting racism, historical trauma, and land and labor exploitation within food systems (Cadieux and Slocum 2015). As Holt-Gimenez (2015) explains, "understanding why, where, and how racism manifests itself in the food system, recognizing it within our movement and our organizations and within ourselves, is not extra work in transforming the food system, it is the work" (p. 2).

Academic research and universities have long been complicit in shaping extractive food systems and have a responsibility to reimagine their role (Grande 2018; Lee and Ahtone 2020). While academic research can offer vital frameworks and resources, it must be accountable to those building alternative and just futures on the ground. Systemic changes toward justice are impossible when research is divorced from the organizing efforts of farmers, workers, and community members (Wittman et al. 2010). Participatory Action Research (PAR) aligns research with community priorities by centering those most affected by injustice and fostering co-produced knowledge and collective action (Fals-Borda and Rahman 1991; Brydon-Miller et al. 2003; Swords 2019; Christens et al. 2016).

In this paper, we present a case study of a multi-year participatory research project in the Upper Midwest, USA, conducted as part of Grassland 2.0 and funded by the US Department of Agriculture, to support transitions toward perennial, grass-based animal agriculture. In the U.S. Midwest, calls to shift away from extractive monocultures of corn and soybean systems that sustain CAFOs and degrade soil, water, and biodiversity have reinvigorated interest in well-managed rotational grazing (Altieri et al. 2015; Horrigan et al. 2002; Rasmussen et al. 2018). Well-managed rotational grazing, in conjunction with conservation grasslands, presents an important alternative to these monoculture-driven systems, as it can stabilize soil carbon, reduce soil erosion, and support rural economies by reducing the reliance on costly inputs (Franzluebbers et al. 2012; Garnett et al. 2017; Spratt et al. 2021). At the same time, global sustainability scholarship (such as Herzon et al. 2024) reminds us that living within planetary boundaries requires improving livestock systems as well as the importance of downsizing meat consumption. In this context, well-managed grazing, when ecologically and socially contextualized, can play an important role in food-climate transformations. Yet, transitions to managed grazing and other perennial systems do not inherently ensure justice. University and academic research that purports agricultural solutions solely based on specific "practices" or "innovations" with blinders to the economic and social inequities embedded in those practices will ultimately fall short of supporting meaningful change. Through 128 semi-structured interviews and three participatory workshops with a racially diverse group of farmers, nonprofit leaders, and organizers, we explored what it takes to foster meaningful, justice-oriented change within the many overlapping inequities of animal agriculture and transitions to grass-based perennial practices. Adapting PAR methods, we incorporated tools from community organizing to engage deeply with participants and foster spaces for dialogue across difference. In doing so, we propose a Just Transitions PAR Implementation Framework that highlights both the conceptual importance of just transitions in agriculture and the methodological interventions and tools that researchers can use to co-produce transformative change. This framework offers a conceptual contribution to the PAR literature and practical guidance for researchers committed to just transitions in agriculture.

## Conceptual intervention: just transitions, PAR, and university research

### Just transitions in agriculture

Just Transitions (JT) is an evolving concept rooted in labor, environmental, and climate justice movements, particularly in communities of color grappling with the racial inequities produced by extractive industries (Alliance n.d.; Lerner 2010). While early definitions centered on distributive fairness and worker issues, current scholarship recognizes JT as a multidimensional framework encompassing broader governance, ecological justice, and epistemic struggles (Wang and Lo 2021; Welton and Eisen 2019). Across these diverse lines, JT highlights the need to restructure systems so that benefits and burdens are shared fairly, historically marginalized communities are recognized, and harms are repaired. In this paper, we follow the line of decolonial and movement-based scholarship that positions JT not just as a policy framework but as a transformative praxis emerging from grassroots organizing and relational world-making, foregrounding racial, decolonial, and multispecies justice. Scholars and movement leaders alike have noted the limits of frameworks rooted solely in Western theories of justice, which often remain anthropocentric and overly focused on distributive policies. Critics argue that such perspectives risk reducing JT to technocratic fixes (Sovacool 2021; Swilling 2020; Williams and Doyon 2019). A growing body of work calls for relational approaches that center interdependence among human and beyond-human communities (Escobar 2015; Fochesatto 2025; Kimmerer 2013; Tsing 2015). Through this perspective, justice is not only a matter of redistribution or recognition but an ongoing cultural and ecological practice of care, reciprocity, and long-term

transformation. Thus, JT is prefigurative in nature (Newell and Mulvaney 2013; Yates 2015) as communities work to enact alternative ways of farming, organizing, and relating in the present, modeling the just futures they seek to build.

From this vantage point, agriculture emerges as a pivotal site for JT (Blattner 2020; Fochesatto 2025). Agriculture, and animal agriculture in particular, concentrates multiple intersecting injustices, from land dispossession to racialized labor exploitation, environmental degradation, and the commodification of animals, while also embodying possibilities for more just and relational futures (Blanchette 2020; Blattner 2020; Fochesatto 2025; Klassen et al. 2023). Approaching agriculture through JT makes visible the need for transitions that are not only ecologically sustainable but that also foreground who is recognized as a farmer, who has access to land, and how multispecies communities are supported in the process.

Despite its broad relevance, JT scholarship has focused primarily on the energy sector, overlooking animal agriculture's central role in climate issues (Blattner 2020; Murphy et al. 2022; Harrahill et al. 2023). While emergent JT scholarship is recentering food systems (see Kaljonen et al. 2023), these efforts rarely engage the racial justice and decolonial critiques that have long been central to food justice scholarship. Dominant livestock agriculture systems in the U.S. rest on legacies of settler colonialism, Indigenous land dispossession, and racial capitalism, shaping who has access to land, who labors, and whose knowledge counts (Baptist 2014; Guthman 2011; Minkoff-Zern 2019). Although this paper focuses on the U.S. Midwest, similar dynamics of colonial expropriation, racialized labor, and epistemic violence structure industrial agriculture globally.

Before European contact, the Midwestern United States, part of Turtle Island, was a vast tallgrass prairie ecosystem stewarded by Native tribes who maintained a symbiotic relationship with the buffalo, a keystone species in this ecosystem (Isenberg 2000). The arrival of European settlers marked a significant ecological and cultural shift. They systematically exterminated the buffalo and plowed the prairies, using these actions as tools of genocide to displace Indigenous people and facilitate colonial conquest and settlement (Hubbard 2014). Subsequently, land allotment policies fragmented these expansive lands into smaller, privately owned plots, enforcing Eurocentric and heteropatriarchal family structures as the normative model for landholding (Justice and O'Brien 2022). This restructuring served as a tool of racial control, marginalizing Indigenous governance and kinship systems, excluding communal land use practices, and laying the groundwork for racialized disparities in land access and resources that persist in today's food systems. Family farms, while diversified in their cropping, struggled with overgrazing due to the limited size of plots, making it difficult to maintain the natural grassland ecosystems (Holleman 2017). Moreover, the promise of prosperity for settlers was often oversold, and many families lost their farms while land speculators and railroads profited (Gilbert 2015)

The colonial legacy of land dispossession was further entrenched by the Green Revolution and the increased industrialization of farming, which has promoted heavily mechanized large-scale corn and soy monocultures at the expense of small and medium sized farms and agroecological practices such as well-managed grazing. This shift has resulted in soil degradation, loss of biodiversity, and increased reliance on chemical fertilizers and pesticides which negatively affect water quality, pollinators, soil carbon, and surrounding ecosystems (Altieri 1999; Gillon et al. 2016; Horrigan et al. 2002; Rissman et al. 2023). The environmental impacts are worsening in regions dominated by industrial animal agriculture, such as the Midwestern US, where confinement animal feeding operations contribute to air and water pollution, including algal blooms and the dead zone in the Gulf of Mexico (Burkholder et al. 2007).

Increased land and market monopolization has caused an exodus of small and medium-sized farm operations and a lack of regional supply chain infrastructure to support local food systems (Hendrickson et al. 2020). For instance, the number of dairy farms in Wisconsin decreased from 24,065 in 1997 to 6,216 in 2022 (USDA NASS 2022). Through systematic policies of dispossession and discrimination, 97% of farmland today is owned by white farm owners in the U.S., while the majority of farm labor and food processing is low-paid exploitative work performed by people of color (Horst and Marion 2019). Increased market consolidation is contributing to skyrocketing land costs that restrict land access to beginning farmers, especially would-be farmers of color who are less likely to inherit family land or attain access to the capital needed to purchase land (Penniman 2018).

Workers in animal agricultural operations face some of the most hazardous working conditions in food work (Fox et al. 2017). Exposure to toxic chemicals, such as pesticides and ammonia from manure pits, can lead to chronic respiratory issues, skin infections, and even fatal accidents (Fox et al. 2017). The reliance on a predominantly immigrant, unauthorized workforce exacerbates their vulnerability to wage theft, physical abuse, and lack of legal protections, further entrenching economic and racial disparities in agriculture (Harrison and Lloyd 2012).

These intertwined histories of dispossession, industrialization, and exploitation, reveal how transitions in U.S. agriculture are deeply entangled with structures of racial capitalism and settler colonialism. Approaches that frame transition as only technical or environmental risk reproducing these inequities by ignoring how power, culture, and

ecology are bound together. Relational and prefigurative approaches in JT center repair and the needed reworking of relationships between peoples, humans and non-humans, and between past harms and future possibilities. In this way, agricultural transitions must do the relational work of grappling with current hierarchies. In what follows, we turn to PAR as an approach that aligns with this by centering the voices of those most affected and by creating spaces where more just agricultural futures can be prefigured in practice.

### Why PAR can support just transitions in agriculture

PAR is well-positioned and conceptually aligned with movement work in Just Transitions because it centers the knowledge and leadership of those most affected by injustice. Rooted in the work of Kurt Lewin, Paulo Freire, and Latin American resistance movements to extractive development, PAR emerged as a collaborative method that integrates research, action, and reflection in the pursuit of structural change (Freire 1970, Leal, 2007; Escobar 1995; Christens et al. 2016). Paulo Freire advocated for research and pedagogy rooted within and conducted by communities to raise a "critical consciousness," or in other words, an understanding of one's social and political situation through a process of inquiry, reflection, and action (Freire 1970). Kurt Lewin introduced the term "action research" to describe the process of studying a social problem to create social change and emphasized that so-called "minority" problems were indeed whole society problems that deserved the attention of researchers (Adelman 1993).

PAR in agricultural research traces back to New Deal-era rural planning and evolved through models like Farmer Participatory Research which sought to integrate farmer knowledge and traditional ecological knowledge into scientific inquiry (Gilbert 2015; Peeters 2015; Bentley 1994). Although these models were transformative for their time, representing a shift away from top-down research, limitations such as superficial engagement and underrepresentation of marginalized voices highlighted the need for deeper, more inclusive processes.

Today, PAR approaches seek to deepen participation, co-design methods, and explicitly address power issues. PAR can be "a testing ground for the utility of theory, new methods in social science, and for new combinations of ideas and methods from various disciplines and fields of practice" (Christens et al. 2016, p. 250). It can also focus on systems-level analyses of complex problems and help communities who have been excluded to finally participate in agenda-setting processes and changing dominant institutions (Christens et al. 2016, p. 244). Méndez et al. (2017) emphasize the alignment of PAR with agroecological principles of collective action, long-term transformation, and structural critique. Rios (2022) argues that researchers must engage in horizontal dialogue with people who are not institutionally credentialed but who possess experiential expertise grounded in social struggle. Hammelman et al. (2020) further stress that PAR must center racial and social equity, particularly in food systems governance, and move beyond reformist approaches to address root causes of injustice.

Yet PAR has its limitations. It is shaped by institutional constraints, uneven power dynamics, and the positionality of the researchers involved that affects the orientation and engagement of research projects. In universities, the dominant metrics of success such as grants, peer-reviewed publications, and high productivity, rarely reward the slow, relational, and sometimes unpredictable work of participatory research (Christens et al. 2016). This reality demands methodological adaptations that are intentional, transparent, and accountable to communities, especially when working in contested fields like agriculture.

## Universities as sites of tension and possibility in just transitions

Universities are powerful institutions in food systems, not just as sites of research but as gatekeepers of knowledge, funding, and legitimacy. Their involvement in food system transitions can either reproduce extractive models or contribute to transformative change. While these institutions can serve as crucial hubs for promoting interdisciplinary collaboration and supporting efforts that move beyond reformist approaches, they also carry histories of exclusion and extraction that must be acknowledged and addressed; from their entanglements in slavery, colonialism, and land dispossession (Grande 2018; Lee and Ahtone 2020; Nash 2019) to the ongoing reproduction of racial and epistemic violence in knowledge production (Spivak 1988; Smith 1999). As Hammelman et al. (2020) emphasize, academic institutions can play a pivotal role in creating more equitable and sustainable food systems by engaging in praxis that links theory, action, and critical reflection. Through collaborations with social movements, civil society, and policymakers, they can help address the structural roots of injustice, including white supremacy, settler colonialism, and capitalist exploitation. PAR within universities must grapple with questions such as who benefits from the research, whose voices shape the agenda, and how to structure participation, compensation, and accountability.

Amidst these tensions, existing literature highlights the potential of PAR to support food justice efforts by bridging the spaces of movement, practice, and research (Hall 2011; Wezel et al. 2009; Adelle 2019). PAR has also been recognized as a vehicle for critical pedagogy and institutional

transformation (Freire 1970; Reynolds & Cohen 2016). While the literature offers strong theoretical alignments between PAR and food justice, engagement with Just Transitions in this space remains emergent and underdeveloped. Further, few participatory case studies examine how university-based researchers navigate the implementation of PAR methods at this juncture. These discussions also highlight the need to clarify the conceptual connections between JT, PAR, and the university. While a JT lens on agriculture exposes the intertwined histories of injustice, PAR offers methodological commitments toward structural change. Universities, meanwhile, embody both extractive histories and potential for change. Thinking through this intersection clarifies the stakes of who is included in transitions, what forms of knowledge and justice are prioritized, and how processes of change are conceptualized and organized. In Table 1, we present this conceptual intervention, synthesizing JT, PAR, and the role of the university.

Through this conceptual intervention, we argue that advancing JT in agriculture from a university research perspective requires pairing JT principles with PAR methodologies, positioning them as mutually reinforcing approaches for systemic change. In what follows, we ground these concepts in an in-depth case study, and building from those learnings, we propose a JT-PAR Implementation Framework to guide agricultural research.

**Table 1** Conceptual intervention: core tenets of jt and the roles of par and the university

| Just Transitions (JT) | Participatory Action Research (PAR) | University and Academic Researchers |
|---|---|---|
| Who: Centers historically marginalized and racialized farmers, workers, and Indigenous communities; asks who has land; who labors, and whose knowledge counts | Who: Centers the knowledge and leadership of those most affected by injustice; builds collective agency and critical consciousness | Who: Can reproduce histories of exclusion or confront who benefits from research, who shapes agendas, and how participation and compensation are structured |
| What: Pursues ecological repair together with racial, land, and labor justice; challenges colonialism, racial capitalism, and industrial exploitation | What: Integrates research, action, and reflection in pursuit of structural change; democratizes knowledge production | What: Can link theory, practice, and reflection; has resources to support transformative change but can also reproduce extractive models if unchecked |
| How: Advances relational, prefigurative practices that create just futures now through care, reciprocity, and long-term transformation | How: Uses horizontal dialogue, co-design, and iterative reflection to shift power and build collective action | How: Foster long-term collaborations with movements, civil society, and policymakers to support systemic change and bridge silos, while practicing humility and de-centering academic control. |

## Case study: grassland 2.0

Grassland 2.0, launched in 2019 with funding from the U.S. Department of Agriculture, was designed to "support an agroecological transformation" in the Upper Midwest through interdisciplinary, community-based research. The focus of the project was on management-intensive grazing, which is receiving more attention as a solution to environmental issues in animal agriculture. In this case study, we show that while grazing is often framed purely in ecological or technical terms, especially coming from the university, questions of racial justice and equity are central to transitions in animal agriculture.

In management-intensive grazing systems, livestock are rotated through a series of paddocks, allowing pasture time to recover and regrow (Oates et al. 2011; Paine et al. 1999). Overstocking animals on pastures can lead to negative effects on soil and water quality (Bilotta et al. 2007), but well-managed grazing has been shown to optimize pasture productivity and enhance soil fertility by maintaining ground cover year-round, preventing erosion and promoting soil health (Wepking et al. 2022; Spratt et al. 2021). Some research indicates that diminished input requirements boost farm profitability, decrease farmers' reliance on debt, and lessen the necessity for scaling up operations to maintain financial viability (Spratt et al. 2021). However, the transition to managed grazing and other perennial systems raises questions about labor conditions and social hierarchies. While there are ecological benefits, these systems do not automatically advance social justice. Furthermore, even community-based research approaches in agriculture can overlook critical power dynamics and inequities.

This project sought to engage researchers, Extension professionals, farmers, and key organizations within the sustainable grazing community. A "meta-stakeholder group," composed of agricultural professionals and farmers engaged in grazing operations, guided these efforts. While decades of research had primarily focused on the agronomics of grazing, the project aimed to address what many farmers saw as the main problem: the economic, social, and political dimensions of grazing. In this way, the transdisciplinary approach, which brought together diverse knowledge holders to address these issues, aligned with some but not all principles of participatory action research (PAR).

In 2020, during the second year of the project, the national reckoning with systemic racism sparked by the murder of George Floyd and the continued subjugation and violence against people of color in the U.S. exposed critical gaps in the project's approach. Researchers and community members alike were compelled to confront the limitations of colorblind approaches and the absence of meaningful engagement with race in their work. Early engagement

efforts had predominantly involved white-led organizations and white farmers, highlighting a broader issue of racial exclusion well-documented across the sustainable agriculture movement (Guthman 2011; Slocum 2006). Thus, the team grappled with whether and how to adapt approaches to better address the linked social, racial, and economic injustices in grassland agriculture.

As part of the project's policy and governance sub-team, we encountered both openings and tensions. While some project members acknowledged the importance of equity, colorblind assumptions often shaped discussions, leading to surface-level engagement with racial justice. Sub-teams sometimes gestured toward inclusion but did not meaningfully adapt methods or theories that could address inequities in agriculture or engage communities of color in their goals. These dynamics were not unique to our case but mirrored a broader pattern of colorblindness and white logics. At times, this manifested itself in dismissive attitudes that, as Holt-Giménez (2015) points out, framed conversations about racism as "extra work" rather than central to agricultural transformation. It was in this context that we turned to JT and PAR. JT provided a way to insist that equity and justice are central to agricultural transformation, not optional, and to recenter our approach around cultural and relational change, attention to racial capitalism, and the valuation of farm and food work. We turned to PAR as a methodological intervention to course-correct. This shaped not only who we engaged but also how we structured the research: purposive sampling to reach underrepresented farmers, stipends to address inequities in labor, and one-on-one conversations as tools for redistributing voice and shaping agendas. These were not neutral methods but direct strategies to resist reproducing the exclusions we observed within the project and the field at large.

Our goal was to confront the project's limited engagement with diverse audiences and identify barriers and opportunities for a socially and racially just transition away from row-crop–dependent animal agriculture toward sustainable, ancestral practices on perennial pastures. We grounded this approach in the understanding that when transitions fail to address inequities and historical legacies of oppression, they risk reproducing those very structures (Gadsden et al. 2023; Holt-Giménez and Harper 2016).

Figure 1 illustrates our project timeline, detailing the major phases and activities we implemented. We began conducting semi-structured interviews in the fall of 2020. These 1-to-2-hour interviews explored participants' experiences with managed grazing, their visions for a just food system, and the challenges, opportunities, and policy and governance issues they faced or observed in their communities. Participants were primarily based in Wisconsin, Minnesota, Illinois, Michigan, and Iowa, except for a few interviewees from non-Midwest states, whom we interviewed to cover specific gaps in perspectives. The majority of participants worked in grass-based agriculture, but we did include several interviewees with farmers and professionals working with conventional crops or confined animals. Further, we built relationships through "one-on-ones" where we held more informal meetings to build trust, shape the research direction, and brainstorm collaborations and methods. These meetings helped establish trust, refine our research direction, and revealed the need for an explicit organizing space around JT in Midwest agriculture. They also led to the co-design of three online participatory workshops held in early 2022, where farmers, nonprofit professionals, and Tribal representatives analyzed interview data, discussed racial equity issues, and developed collective strategies.

Table 2 outlines the goals of the workshops, which focused on evaluating early findings from the 128 interviewees and using them as a foundation to address justice issues in the sector. The structure consisted of three workshops, spaced two to three weeks apart and lasting approximately four hours each. Before the first workshop, we employed a blended approach of inductive and deductive coding of the semi-structured interviews (in NVivo), developing deductive codes from core concepts in the interview guide and recurring issues participants mentioned (Saldaña 2021). We synthesized emergent themes to present the data in a clear and accessible format to workshop participants. We originally planned to include in-person sessions for better relationship-building opportunities, but due to the Omicron COVID-19 variant, we had to shift to an online format. Participants included six farmers, thirteen non-profit professionals, four individuals from Tribal organizations, and two representatives from educational institutions. Farmers received $40/hour, and NGO participants received a $100 stipend. Providing compensation was essential in ensuring a diverse group of community members could participate.

We built a significant amount of flexibility into the workshops to allow for a continuing process of co-design. Evaluation surveys after each workshop informed changes to facilitation strategies as well as the activities for the next workshop. Each workshop built upon the last. Workshop 1 focused on visioning just transitions, Workshop 2 identified collective actions, and Workshop 3 explored strategies for building collective power.

Past the workshops, participants asked us for a summary of workshop themes and recommendations. We published a report and six briefs on the project website in January 2023. The open-access report offered a timely, accessible alternative to peer-reviewed publications. To share findings and engage participants, we hosted a webinar featuring researchers and farmers discussing the PAR process, key

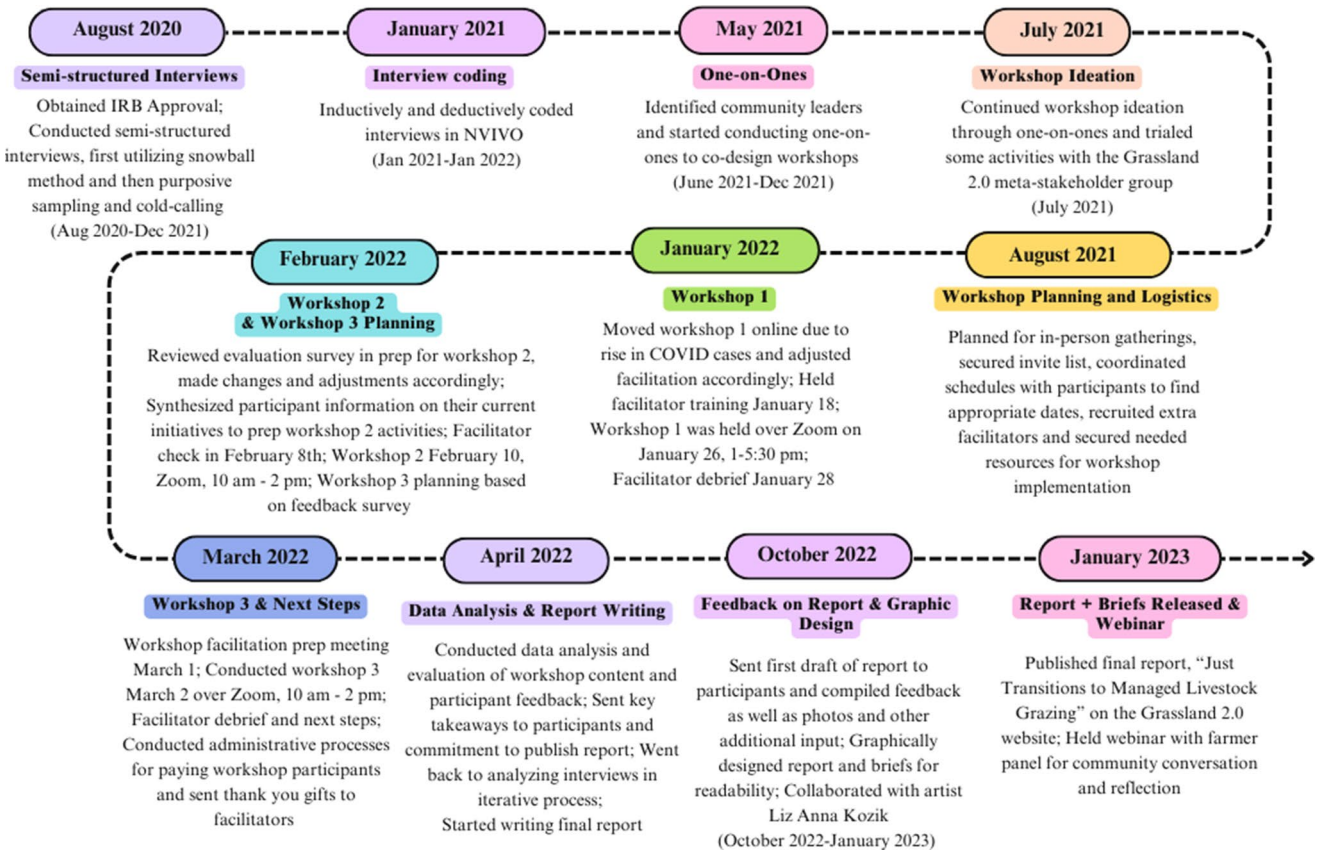



**Fig. 1** Step-by-Step Project Timeline

**Table 2** Workshop structure, goals, and activities

| Workshops | Goals | Agenda (see Appendix A for details) |
|---|---|---|
| **Workshop 1: Visions for the Future** | - Share qualitative data, initial analytical themes, and co-analyze "just transition" to grassland agriculture.<br>- Envision what a just transition might look like and lay the groundwork for action planning and collective power building. | - Icebreaker and introductions over lunch.<br>- Presentation by project Principal Investigator.<br>- Overview of just transitions definitions and interview themes.<br>- Set discussion guidelines.<br>- Visioning justice in agricultural transitions through multiple rounds of a World Café activity, with breaks and reflection periods.<br>- Wrap-up and preparation for Workshop 2.<br>- Participants are asked to fill out a form to detail their current initiatives and how they align with just transitions. |
| **Workshop 2: Actions for Transition** | - Revisit visions from the first workshop and assess current work and initiatives among participants.<br>- Identify collective strengths and opportunities for collaboration and action towards just transitions. | - Welcome and icebreakers to re-engage participants.<br>- Recap of visions from Workshop 1.<br>- "Who is doing what?" Exercise. Small group discussions to share current and aspirational work related to just transitions.<br>- Quiet reflection and discussions on collective strengths and gaps in current initiatives.<br>- Presentation on needs and opportunities for change collected from interviews, followed by discussions in small groups.<br>- Issue-specific discussions and preparation for Workshop 3. |
| **Workshop 3: Building Collective Power** | - Build collective power to bring about actions identified in Workshop 2.<br>- Identify ways to organize more effectively both within and outside existing structures to support the just transition. | - Recap of previous workshops and introduction to building collective power.<br>- Quiet reflection on effective organizing and strategies.<br>- Discussions on organizing tools, scarcity mentality, and power dynamics.<br>- Issue-focused discussions based on prior workshops' insights.<br>- Mini-workshops on storytelling, cooperative structures, and farm bill policy advocacy.<br>- Final wrap-up and feedback session. |

recommendations, and their relevance to farmers, farmworkers, and Tribal nations.

By engaging PAR's methodological focus on power, we redirected the research to address exclusions in the original project's design, center justice, and generate spaces for collective visioning and action. The analysis that follows draws on concrete instances that illustrate how PAR was practiced at the intersection of JT and the university. In the following sections, we analyze the four key elements we learned from this approach: engagement and trust-building, collaboration and co-design, action and co-production of knowledge, and reflection and iteration. These elements, included later in Fig. 3, form the foundation of the implementation framework we derive from this case. Together, we offer them as a guide for how PAR can be used as a method to implement just transitions in agricultural research.

## Engagement and trust building

Trust can never be assumed, and certainly not in agricultural research where communities of color have been historically marginalized and excluded, particularly by land-grant universities and agri-food organizations. Often, in this context, distrust is a relational response to histories of dispossession, racism, and extractive research practices. PAR scholars emphasize that trust is both a process and an outcome of equitable engagement, requiring reciprocity, transparency, and long-term relationship-building rather than one-off recruitment (Cuéllar-Padilla and Calle-Collado 2011; Christens et al. 2016; Hardie Hale et al. 2022). We approached trust-building, therefore, not as a precondition but as a methodological and political commitment central to JT and PAR. Moving beyond established academic and privileged networks required intentional strategies and sustained commitment to outreach, particularly with underrepresented groups. This process expanded participation and uncovered significant insights that shaped the research.

Initially, the research team employed a snowball sampling method, leveraging existing connections within the project (Biernacki and Waldorf 1981). However, a common problem with this method is that it is not well-suited to connect with hard-to-reach populations such as farmers of color and farmworkers (Erickson 1979). Recognizing that this narrowed who counted as legitimate participants, we shifted to purposive sampling to extend participation to those structurally excluded from dominant grazing spaces, allowing us to go beyond the 'usual suspects' or the typically more powerful groups (Guthman 2008; Nordling and Pugh 2019). We actively sought farmers of color, Tribal producers, workers, and organizations supporting racial justice in agriculture by conducting cold calls and cold emails and providing stipends that compensated for participant time and expertise. Outreach time and efforts then focused on specific referrals from those groups. These strategies expanded engagement, while deeper trust-building required more sustained, relational practices.

To cultivate relationships and build trust, we borrowed methods from community organizing, specifically one-on-ones (Stoecker and Falcón 2022). These informal dialogues emphasized transparency and reciprocity over data extraction. We shared our intentions openly, asking whether and how our research could be useful to grassroots organizing efforts, and acted on the feedback we heard. Re-engaging people we had previously interviewed helped nurture longer-term relationships and demonstrated consistency between our words and actions. Through these exchanges, participants began to view the project as collaborative rather than extractive, setting the groundwork for co-design. This ethic of reciprocity also shaped facilitation as we structured workshops around participant concerns and continually adapted based on their suggestions. These consistent acts of transparency, responsiveness, and follow-through became the core of our trust-building practice.

We saw evidence that trust was growing as participants shared frustrations with existing systems and invited us into organizing spaces. These strategies became direct interventions aimed at redistributing voice and countering the distrust that many organizers, people of color, and Tribal producers reasonably hold toward academic research. At the same time, building and maintaining these relationships required extensive time and flexibility, and even then, gaps remained. For example, although we interviewed some livestock farmworkers, they did not make up a substantial portion of the interview set or participate in other parts of the project. We struggled to find and engage farmworker organizers, a gap that reflects the broader lack of organizing capacity around livestock farmworkers in our region at the time of this research. While organizations such as Voces de la Frontera and the Farmworker Project provide critical support for farmworkers generally, initiatives specifically focused on livestock labor organizing were limited. Promising models elsewhere, such as Vermont's Migrant Justice and the Milk with Dignity Campaign (Justice n.d.), demonstrate the potential for better support for livestock workers. The intensive nature of our participatory workshops, combined with farmworkers' long hours, further limited participation, though contributions from a university researcher with expertise on livestock labor (Ibarra et al. 2023) brought important perspectives into the process. The lack of deeper engagement with farmworkers remains a key limitation of this case, underscoring that trust-building and engagement are always partial and ongoing and that future efforts require more targeted strategies for bridging silos within labor and agriculture.

Even so, what emerged through these imperfect but sustained efforts was a broader and more diverse set of relationships than typically seen in grazing research. The final sample included 22 participants of color (17%), with an additional 13% working in organizations led by or serving people of color. 40% were farmers (with a minority representing farmworkers and aspiring farmers); 25% were affiliated with nonprofit organizations; 14% worked for the U.S. government; 10% were involved in Tribal governments (mainly in Wisconsin); 6% were in educational institutions; and 5% were in other sectors, such as financial institutions (for more details on sample characteristics see Appendix B). The racial makeup of the sample is a significant departure from the demographics of Midwest agriculture, where USDA data show only 2% of farms are operated by Black producers and 3% by Hispanic producers (USDA NASS 2022; Horst and Marion 2019). We often encountered claims from other researchers and white farmers that "few farmers of color are involved in animal agriculture," but interviews and one-on-ones revealed that many farmers of color are raising chickens, goats, and pigs, even if structural barriers around land and capital access have made entry into beef and dairy more difficult (Penniman 2018). Farmworkers who are disproportionately people of color (Holmes 2013; Harrison and Lloyd 2012) should be included in JT efforts even if they do not currently have land titles or a voice in farm management. Tribal partners also emphasized the cultural and ecological importance of buffalo grazing (Isenberg 2000; Hubbard 2014), expanding the frame of "perennial livestock" beyond what mainstream grazing networks typically consider. These insights not only diversified participation but also directly challenged dominant narratives about grazing and who gets to be recognized as a farmer.

## Collaboration and co-design

Traditional academic research often struggles to support authentic community collaboration, frequently perpetuating power imbalances rather than addressing community needs (Cuéllar-Padilla and Calle-Collado 2011; Rios 2022). A persistent critique is that researchers collect information from communities without reporting back results (Smith 1999). JT challenges this model by centering equity and justice as central to agricultural transformation, while PAR offers the methodological commitments, such as centering grassroots expertise and shared agenda-setting, that make that recentering possible from the university vantage point (Huntington 2000; Peeters, 2019). In this case, collaboration was not simply about including participants in pre-designed activities, it involved co-designing the process itself, with participants shaping the content, pace, and goals of research.

The one-on-one conversations that began during the engagement phase became a diagnostic tool for co-design. These meetings allowed us to understand the personal motivations and networks of community leaders, and generated direct input on how the project could better support transitions for historically excluded farmers and workers (Christens et al. 2016; Braun et al. 2022). Several concrete insights emerged. First, participants stressed the danger of creating yet another coalition in the Midwest sustainable agriculture space. They pointed out that many organizations already existed, but their work was siloed, with limited coordination across farmer, Tribe, and NGO efforts. In response, we used our university role to help bridge existing coalitions and reduce duplication, rather than starting something new. And although a regional coalition was not recommended, this process did ultimately create the conditions for the formation of the Wisconsin Grazing Coalition[1], which filled an identified need for Wisconsin state-level policy coordination without replicating existing efforts.

Second, participants consistently emphasized the importance of solution-focused dialogue. For example, several organizers explained that repeated exercises centered on "Problem Analysis" (O'Hara 2009) often left community members feeling demoralized and fatigued. Similarly, members of the project's meta-stakeholder group warned that too much emphasis on barriers risked alienating participants who already felt excluded from mainstream agriculture. Thus, we structured workshops around actionable and inspiring discussions: envisioning new land tenure arrangements, sharing strategies for de-commodifying agricultural markets, and identifying cooperative structures that could support farmers of color. These concrete activities balanced acknowledgment of systemic racism and structural barriers with attention to hope, agency, and the possibility of change.

Third, community leaders named explicit priorities for the workshop space: to interpret interview data collectively, to center racial equity explicitly, to build collective power, and to move beyond individualized "farm-level fixes" toward broader food systems transformation. For example, discussions with Tribe representatives emphasized buffalo grazing as a site of both ecological and cultural restoration, while conversations with Latinx farmers highlighted the need for bilingual technical assistance and cooperative marketing to overcome structural exclusion. These priorities underscored that collaboration could not be symbolic but that it had to reshape the research questions and methods themselves.

Finally, participants anticipated that the project would involve difficult conversations and explicitly requested skilled facilitation. To meet this need, we recruited experienced facilitators from Extension and the broader project team. This helped ensure group facilitation skills and aimed

[1] Information about the grazing coalition can be found online: https://www.michaelfields.org/grazing.

to foster lasting relationships between community members and university-based agricultural professionals. Agricultural professionals, including in Extension, are rarely trained to facilitate conversations about race, power, and justice. Although most facilitators in this case had some experience engaging immigrant communities and communities of color, few had depth of training necessary to navigate difficult, racial justice focused dialogue. This became evident in one breakout group, where an interaction between a white facilitator with little racial justice training and a white participant seeking to push the conversation further was mishandled and ultimately reported by our research team to the Institutional Review Board (IRB). Without adequate training and experience, well-intentioned professionals may inadvertently reinforce exclusions when they lack the tools to recognize or respond to racial dynamics. This highlights the critical need for comprehensive racial justice facilitation training within agricultural institutions, while also recognizing that difficult moments are inevitable when equity and justice are centered (Akom 2011; Matias 2016).

Further, our collaboration surfaced other challenges familiar to community-based researchers working within university structures. Processing honoraria for participants was difficult and cumbersome, requiring many weeks of coordination with financial university staff and the completion of tax forms for small amounts of money at or below $250. This issue highlights the administrative inefficiencies at universities that continue to hinder community-engaged research, and that must be addressed at an institutional level to scale up these collaborations.

Our positionality as researchers also shaped possibilities in this case and especially for collaboration and co-design. As younger, early- to mid-career women in a male-dominated agricultural field, we sometimes encountered barriers to collaboration. Yet these identities also enabled us to move beyond entrenched grooves of university-community partnerships. The first author's bilingual and bicultural background deepened engagement with Spanish-speaking and Latin American diasporic farmers, while participatory approaches allowed us to bridge gaps between academic and community forms of knowledge. And yet our involvement in academia created an inherent epistemological distance from participants, many of whom have practical, lived experiences in agriculture that differ from our own. Pausing on this distance can be clarifying, as it underscores that while implementation of JT-PAR research does not erase structural inequities, it can provide resources, legitimacy, and convening power to grassroots efforts that can. Co-design and collaboration in this context become less about dissolving the gap between researchers and participants and more about creating spaces for critical consciousness and redirecting institutional privilege toward justice.

## Action and co-production of knowledge

The collaborative creation of knowledge is not neutral or necessarily always positive. It can reproduce epistemologies of social hierarchy or challenge them by centering grassroots expertise and lived experiences (Huntington 2000; Peeters 2019; Rios 2022). JT reminds us that action for change cannot be reduced to technical improvements in grazing systems but that it must be relational and systemic, addressing inequities of race, labor, and land. For us, action and co-production toward that goal meant creating spaces where participants could interpret data, share priorities, and design strategies for systemic change.

The three participatory workshops exemplify this approach. Rather than leading with scientific presentations, as is often the case in university-community events, we deliberately limited researcher input and used interview data to catalyze discussion. Workshop 1, "Visions for the Future", we opened with structured introductions and icebreakers to set a welcoming tone, and shared interview themes in a short presentation (see Lowe et al. 2023). Participants then engaged in a World Café (see Löhr et al. 2020 for instructions on activity facilitation), moving between small groups where facilitators posed questions grounded in the interview data, ranging from visions for agricultural livelihoods to confronting dominant paradigms that perpetuate racial injustice. Questions included "What does it look like for all in agriculture to be valued and supported?" and "How can we create support structures that allow farmers to live, as one farmer interviewee said, 'with the land, rather than off the land'?" Another World Café round focused on issues of democratization, asking "What are some dominant ways of thinking you see in your work that have implications for racial injustice? How could we shift these ways of thinking to address the racial injustice perpetrated by our food system?" One participant later reflected, *"It was great to have the chance to hear about new ways of conceptualizing things—like de-commodifying land … Having concrete examples helped these concepts go from vague ideas to understandable goals, actions, or policies."* This allowed for deep conversations and relationship-building while further analyzing the data and discussing how to apply these insights (Cadieux and Slocum 2015).

Workshops 2 and 3 built on this foundation by moving from vision to action. Participants first mapped ongoing initiatives through a "Who is doing what?" exercise and helped co-create the *Midwest Perennial Animal Agriculture Initiatives Directory*. NGO participants reported using the tool to coordinate across organizations, while farmers noted it helped them locate resources for their operations. We then synthesized interview data into eight categories of systemic challenge: education, alternative markets, processing,

essential workers, consolidation, capital, land access, and social norms (Lowe et al. 2023, p. 6). Figure 2a is a diagram we presented to workshop participants to illustrate these categories and their interconnections. Participants reflected on this diagram and used dot voting (1 dot per person) on the issue they believed was most critical for achieving JT. As shown in Fig. 2b, participants added key issues they felt were missing from the diagram and voted most to prioritize actions related to supporting improving land and capital access and shifting social norms. They then revisited the "Who is doing what?" exercise and critically compared it to the prioritization from the dot voting activity. This exercise revealed a collective recognition that dominant transition efforts in the region seemed to be focused heavily on education and market-building and were misaligned with the structural issues most impeding justice, namely consolidation, access to land and capital, and changing social norms (Lowe et al. 2023, p. 11).

Specific facilitation and co-design principles laid the groundwork for action here by accommodating diverse cognitive and communication styles and through integrating multi-modal strategies: five-minute periods of quiet reflection, visual note-taking, small-group dialogue, and tools like MURAL and later Google Slides. Some participants valued the digital platforms, while others faced technical barriers, particularly with rural broadband. This forced continuous adaptations which are another example of PARs iterative commitment to accessibility and responsiveness. COVID-19 made these online formats necessary, expanding geographic reach across the Upper Midwest but constraining the intimacy of in-person dialogue.

Workshop 3 was intentionally structured as a blank canvas. Drawing on priorities identified earlier, we co-designed the structure in the weeks between Workshop 2 and 3. The product were small-group strategy sessions and mini-workshops on storytelling, cooperative structures, and policy advocacy. The policy mini-workshop was led by a participant, demonstrating how facilitation itself could be redistributed. At the end of the workshop series, there was a collective request for written outputs, leading to an open-access report and six policy briefs. These documents were not conceived as conventional academic products but as tools for advocacy and organizing. Participants explicitly asked for resources they could cite in policy meetings, point to in NRCS offices when seeking support for grazing operations, and use to legitimize the knowledge they had co-produced together. The report and briefs detailed the inequities plaguing grazing communities and distilled recommendations around issues such as consolidation, land access, and capital, providing shared language to press for structural change. In this way, the outputs served as both evidence of collective analysis and instruments for ongoing struggle, extending the reach of the workshops into other institutional arenas and organizing spaces. As one NGO organizer reflected after Workshop 1: *"We already had a meeting with [another organization] to see how our groups can work more together!"* This illustrates how the workshops functioned as catalysts, with participants carrying

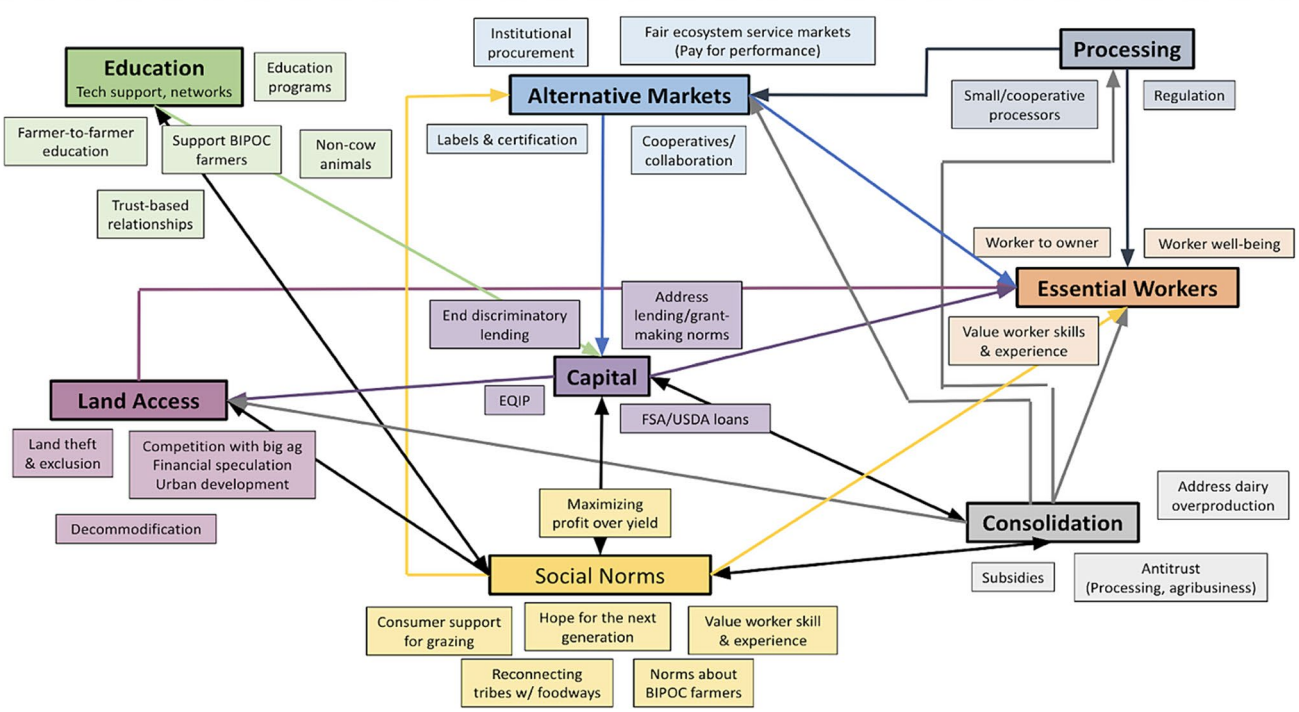



**Fig. 2a** Analysis of needs and opportunities to promote equity and well-managed grazing in agriculture based on interview data.

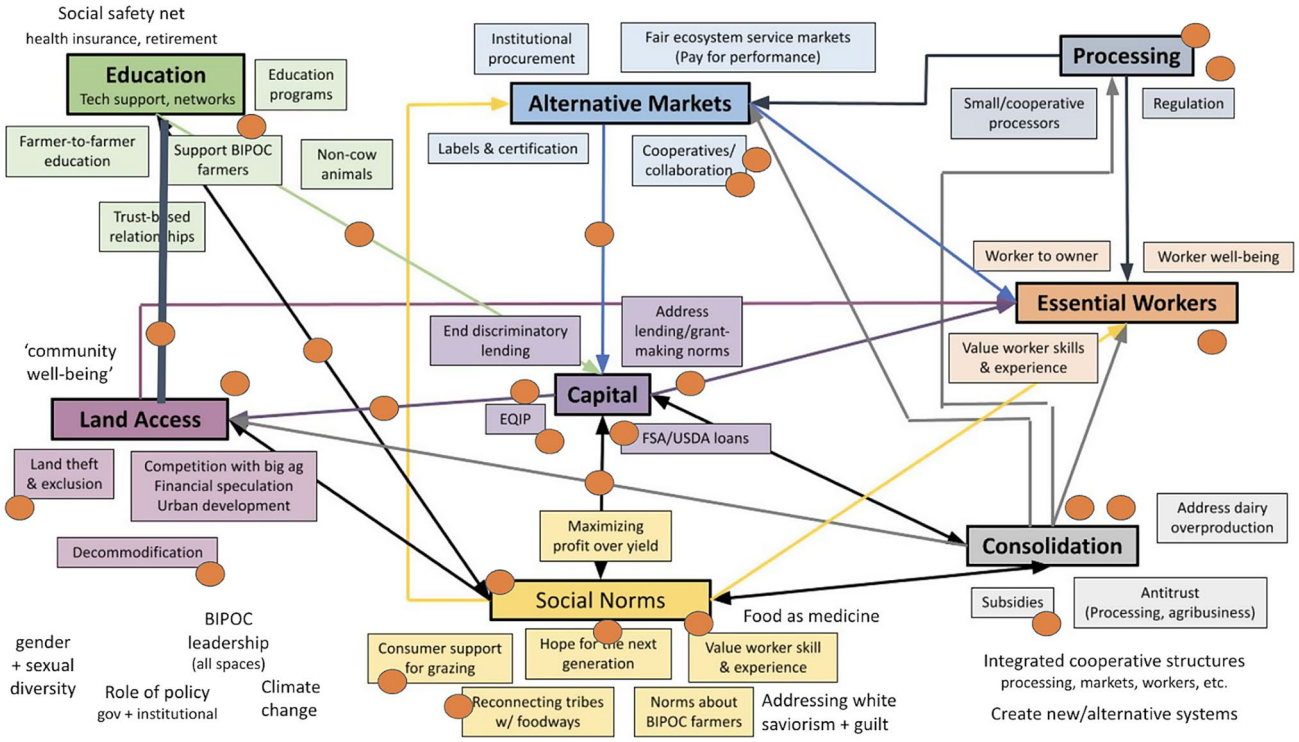



**Fig. 2b** Priority areas for just transitions: workshop participants' dot-voting results highlighting key intervention points

relationships, insights, and now shared documents into their own networks and initiatives.

## Reflection and iteration

Reflection was intentionally built into every part of this project as a critical component of praxis, fostering critical consciousness, surfacing systemic inequities, and enabling iterative adaptation. In JT-PAR terms, reflection was not an add-on but a necessary practice for confronting privilege and positionality, both our own as university researchers and those of participants, while also reworking how relationships and knowledge production unfolded.

Reflective practices were integrated into the workshops through structured quiet periods, rotating group dialogues, and post-workshop surveys that allowed participants and researchers alike to pause, process, and adjust. As one participant explained, *"The most useful part of the workshop was hearing the quotes from the collective interview data. Being able to ruminate over my thoughts and the thoughts of the collective was helpful."* Another emphasized the value of diversity in reflective spaces: *"I really enjoyed getting such a great mix of people and changing up the groups! It was neat to hear from so many voices."* These reflections align with Bain et al. (2021), who argue that incorporating reflection into community dialogues deepens engagement and strengthens critical thinking. Our participants specifically pointed to quiet reflection time as crucial for keeping multi-hour Zoom sessions "very productive," highlighting how structured pauses supported not just participation but also justice-oriented engagement.

Reflection was also distinctly iterative in character. Feedback after Workshops 1 and 2 directly shaped the design of Workshop 3, which was intentionally structured as a "blank canvas" to allow participants to set priorities and lead strategy development. Similarly, flexibility was important in all activities such as in the example of how technical difficulties with online platform MURAL in early workshops prompted us to shift to Google Slides, which was more accessible for participants in rural areas with limited broadband. While this may appear as a minor logistical shift, it exemplifies the importance of reflection and iteration which allow for flexibility to respond to participant needs, ensuring that design choices reflect both equity and accessibility.

Critically, reflection generated deeper reckonings with inequity and privilege. Participants shared candid insights: *"I learned more about the barriers that BIPOC*[2] *farmers in our discussion groups had faced."* Another reflected, *"I'm afraid I learned how much I have to learn about the many things I've taken for granted as a white senior age farmer."* These statements illustrate how reflection opened space for participants to examine their own roles in systems of inequality, embodying Freire's (1970, p. 35) conception of conscientização as "learning to perceive social, political,

[2] BIPOC refers to Black, Indigenous, and People of Color.

and economic contradictions, and to take action against the oppressive elements of reality."

At the same time, reflections from our perspectives expose limits and exclusions. As previously notes, farmworkers were underrepresented, as were stakeholders who were skeptical of racial justice in agriculture. We did not aim to engage these skeptics directly, as our priority was to create safe spaces where an energized core could build momentum. Yet their absence suggests that separate, parallel processes are necessary to build solidarity across constituencies. Reflection also drew our attention to our own institutional constraints within the university and the limits of our capacity.

Although we had USDA funding that supported two full-time graduate students, faculty summer salary, administrative assistance, and volunteer facilitators, we often felt stretched thin and unsupported. Managing administrative hurdles such as the cumbersome process of compensating participants was particularly draining, revealing how existing university systems actively undermine equitable participation. There is a critical need for adequate compensation and institutional support for both graduate students and staff engaged in community-based research. Sustainable support structures are essential not only to address the labor-intensive nature of this work but also to ensure that researchers can fully commit to the complexities of collaborative projects. Systemic changes to academia are necessary to prioritize and fund community-based research (Stoecker 2009), particularly research that aims at transformational change. While the outcomes of these kinds of projects may not always be immediately visible or easily quantifiable for funding agencies, they are nonetheless significant in laying the groundwork for systemic and lasting progress.

Taken together, these practices demonstrate that Just Transitions are not linear or easily measurable outcomes but iterative cycles of dialogue, adaptation, and accountability. For universities, reflection demands humility and vulnerability, including shifting from positions of expertise to roles as facilitators and resource providers for grassroots change. In this way, reflection functions critically for participants and researchers alike as a praxis (in the Freirean sense) that allows for a collective process of naming contradictions, interrogating positionalities, and imagining new forms of action to deepen commitments to justice and reimage the university's role in agricultural transformation.

## Deriving a just transitions PAR Implementation framework

Drawing from the case study, Fig. 3 presents a conceptual model that integrates this paper's contributions to how JT and PAR interact with university research, and the methodological contributions that constitute the JT-PAR Implementation Framework. The figure illustrates how commitments to justice (roots), ongoing movement initiatives (grass), JT-PAR processes (wind), and university research (cattle) are interdependent.

At the foundation are the roots of Just Transitions, which provide the commitments that nourish and guide the ecosystem. These include centering racial justice, confronting historical inequities, challenging extractive systems, expanding networks, and including underrepresented groups. These commitments orient the work toward equity, preventing transitions from reproducing the very injustices they seek to undo. Above ground, the grasses represent the visible, farmer- and community-led initiatives that constitute ongoing movement building: Tribal food sovereignty, nonprofit collaborations, policy advocacy, alternative markets, and farmer-led education. These efforts demonstrate that Just Transitions are already underway, often outside or alongside the university. The grazing cattle symbolize the tensions where university research sits in agricultural transitions. On one hand, they evoke the colonial and extractive history of livestock production in North America, where cattle were instruments of dispossession and enclosure. On the other, they reflect the centrality of well-managed grazing in ecological regeneration. They remind us that cattle, like universities, can either reinforce extractive systems or be in good relations toward ecological and social repair. Universities can reproduce harm through extractive research and siloed agenda-setting, but they also hold resources and convening power that can support coalitions, amplify marginalized voices, and sustain community-led initiatives.

Circulating through and connecting these layers are the JT-PAR processes, represented by the "wind" that pollinates the grasses. Zooming in on the participatory process, Fig. 4 isolates the JT-PAR Implementation Framework from the broader conceptual model to highlight its iterative design. The four overlapping elements we've analyzed in this paper structure the implementation: Engagement & Trust, Collaboration & Co-Design, Action & Knowledge Co-Production, and Reflection & Iteration. These overlapping circles represent the framework not as discrete steps but as iterative, mutually reinforcing practices. Trust enables collaboration; collaboration generates collective action; action demands reflection and iteration; reflection deepens trust and renews collaboration. Many methods we have presented, such as one-on-ones, participatory workshops, or reflective exercises, function across multiple elements at once. This overlapping design emphasizes JT-PAR as the "wind" that directs the herd. Like cattle, universities can either erode landscapes through extractive practices or, when guided by careful stewardship, contribute to ecological and social care. The overlapping bubbles constitute the JT–PAR Implementation Framework, showing how participatory praxis can redirect institutional power toward equity while sustaining the broader movement for JT.

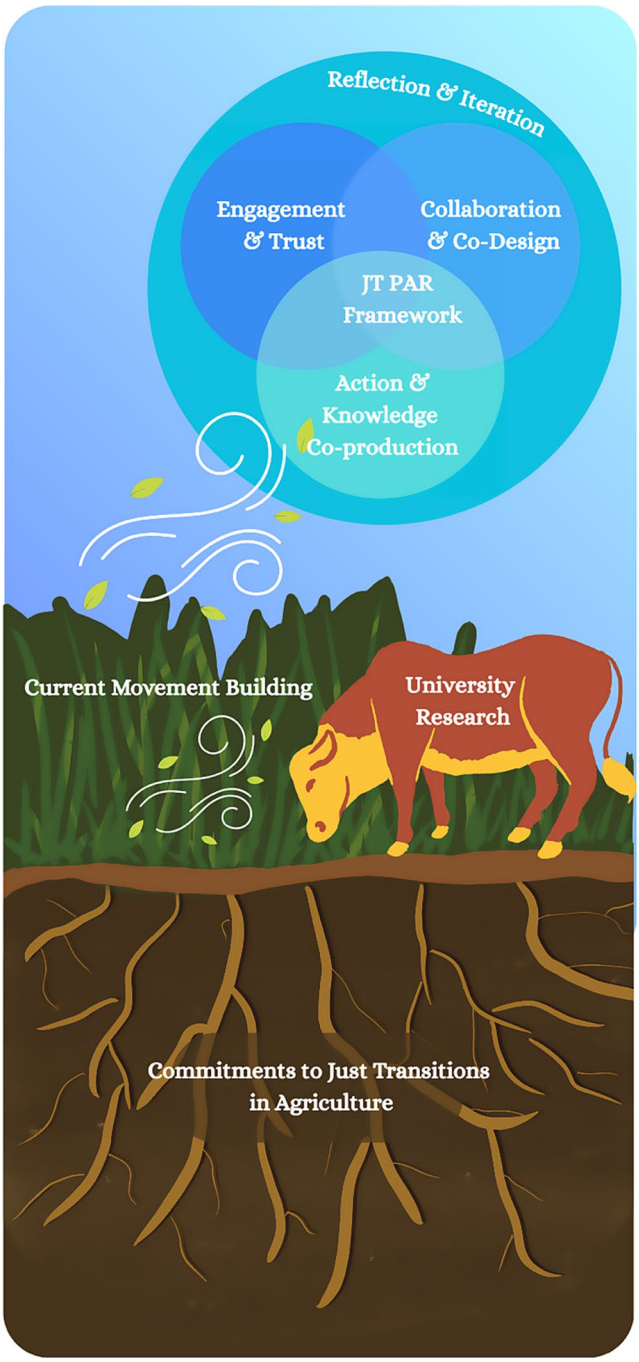



**Fig. 3** *Conceptual model of the university, just transitions, and participatory action research.* this metaphorical landscape depicts how commitments to justice (roots), ongoing movement initiatives (grass), JT-PAR implementation processes (wind), and university research (cattle) interact as an interdependent ecosystem

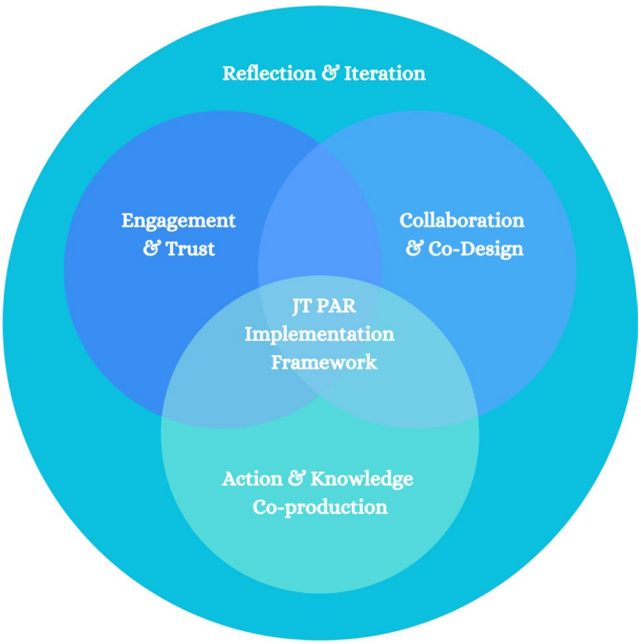



**Fig. 4** *The JT-PAR Implementation framework.* the overlapping circles illustrate how the four iterative processes, engagement & trust, collaboration & Co-design, action & knowledge Co-production, and reflection & iteration, are interdependent and mutually reinforcing. Together, they represent a dynamic, iterative approach to implementing just transitions through participatory action research

## Conclusion

This case study contributes to ongoing debates about the university's role in agricultural transitions, showing how PAR, when re-oriented through a JT lens, can reshape both research and practice. While universities hold resources and technical expertise, they also carry extractive histories and institutional logics that often undermine justice. The lessons from this case study and the framework we propose emphasize that JT-PAR's greatest contribution is in catalyzing ongoing, multi-directional processes of trust, co-design, action, and reflection that redistribute power and sustain collective momentum. As one participant expresses: *"Learning about the work of the organizations involved makes me hopeful for change. We need diversity in agriculture, and in rural areas, the same way it's needed in the rest of society."* Their words express the potential for Just Transitions-PAR at the university to support the ongoing practice of collectively reimagining the world we inhabit, reshaping power, cultivating relationships, and co-creating new realities for an equitable future. Ultimately, transitions are driven by community members through their daily work, organizing, and commitment. The role of researchers and educators is to acknowledge the ways in which their work affects, intersects, and can support these grassroots efforts. Where we route resources, whose knowledge we amplify and engage, and where we direct our attention, all have tangible consequences on advancing justice in transitions.

## Appendix A: Individual reflection and planning guides

These guides were provided to participants via Google Drive, allowing them to download or print the materials as needed. Participants used these guides during Workshops 2 and 3 to support their quiet reflection periods and group Discussions. The guides offered space for note-taking, drawing, and other forms of personal planning and engagement as we moved through each activity.

**Justice in Agricultural Transitions: Workshop 2**

### February 10, 2022

Use this document to help you track your thoughts, observations, and future action items during the course of this workshop. If you print it out ahead of time, you can even doodle on it! This is for your own personal use. We will not be asking you to share.

### Re-visiting visions from workshop 1

What resonates the most with you? What pulls at your heartstrings?

Who's Doing What: Share Your Current and Aspirational work

Share the following with your small group:

1. Name, org/farm
2. What work are you doing at the intersection of perennial animal agriculture and racial equity, consolidation, land access, or cooperative/collaboratives?
3. How does your work connect with/support racial equity efforts and *why* is this important to you?

Note any interesting initiatives you might want to know more about or people you might want to reconnect with:

### Identifying collective strengths & opportunities for collaboration

Reflect on the following questions regarding the initiatives directory:

1. *In what areas are we collectively strong (around what topics are current efforts concentrated)?*
2. *In what ways is the work we're doing supporting the vision for a just transition? In what ways is the work we're doing supporting racial justice?*
3. *Are you planning/hoping to expand your work or add new initiatives in ways that could help support the vision for a just transition? If so, how?*
4. *What is needed to help coordinate current efforts within a particular area (more or different organizing structures, databases, more relationship building, new lines of communication, etc.)?*

What stood out to you from the Discussion?

**Food system mapping: needs & opportunities for change**

### Discussion

Reflect on these questions

- Looking at this food system diagram, what parts of the diagram do you spend most of your time thinking about?
- On what part(s) of the food systems map do you think collective efforts around racial equity should focus? How could you connect your current or aspirational work connect to these anti-racist efforts?
- Are there needs or opportunities you would add to this map to better support racial equity in the food system?

### Issue-specific discussions

Introduce yourself to your group

- Your name, org/farm
- *Why is this issue a priority for you?*
- *How does your current or aspirational work connect with this issue (if at all)?*

Think about ways this group could work together

1. *What are some potential places for collaboration within this group?*
2. *How should our efforts be focused to ensure we center racial justice?*

General reflections from the day

Other questions to think about

- Check out the assets and needs sections at the end of the **initiatives directory**. What are some things others need help in that you think you might be able to support?
- What are some initiatives that you'd like to spread the word about?
- From everything you heard today, what are you most excited about?
- *For organizations:* What are some new initiatives that your organization might be interested in exploring (either expanding current work or exploring new areas)? Are there other organizations that could be potential partners

in this work? Who is the point of contact for those organizations with which collaboration might be useful?

**Justice in Agricultural Transitions: Workshop 3**

## March 2, 2022

Use this document to help you track your thoughts, observations, and future action items during the course of this workshop. If you print it out ahead of time, you can even doodle on it! This is for your own personal use. We will not be asking you to share.

## Workshop 1 and 2 recap

1. What stood out to you from workshop 1 and 2?
2. Any lingering questions or concepts to follow up on?

## Building collective power within and outside existing organizing structures

Framing the discussion

"We need to have a relationship of trust … we've worked with generations of dairy farmers in Wisconsin. You go into the field offices, you'll see great grandpa's casefile in there and we just worked with the great, great-grandson, right. [That level of trust] exists more strongly in our traditional clientele than it does with new groups."

Many interviewees expressed similar sentiments to this USDA employee about the importance of building trusting relationships with BIPOC communities to connect them with resources that other farmers have had access to for a very long time. We wanted to frame the conversation with this quote to emphasize how important trust is to creating more effective organizing structures, supporting BIPOC communities, and distributing resources more equitably.

## Discussion questions

1. What are some of the current regional organizing structures/conveners in the Midwest?

*[Add your response to this question in the chat at the end of the quiet reflection. We'll use it as a reference for the Discussion questions below.]*

2. Share some examples of effective organizing structures or an experience that culminated in effective collaboration.
3. Are there alternative organizing structures or changes to existing structures that would make them more effective in supporting racial justice?
4. What tools could help us organize more effectively? What is needed to maintain these tools?
5. How can we move away from a scarcity mentality to work more collaboratively?

## 6. Power dynamics manifest in funding allocation, collaboration-building, and decision-making power in collaborations. How can we address power dynamics to support BIPOC leadership in regional organizing?Issue-focused discussions

Issue chosen:

## Discussion questions

*[Add your response to all questions in the chat at the end of the quiet reflection]*

1) What are the assets, areas of expertise, or lived experiences you have that relate to this issue (e.g., funding sources, relationships, education programs, policy-advocacy expertise, family/personal history)?
2) Identify 1–3 actions you hope to take or topics you'd like to learn about to help further racial justice.
3) Do you want or need additional collaborative partnerships to achieve this action?Action planning notes

What actions did your group choose to focus on?

What are some of your main takeaways?

What are your next steps (e.g., action items, relationships to build, additional skills/assets needed)?

What connections can be made between this issue and other movements or areas of work e.g., climate change, health, labor, environmental justice, education, food distribution/purchasing?

## Discussion Guidelines:

- **Share, even if you don't have the right words**
- **Be comfortable with silences**
- **W.A.I.T – "Why Am I (still) Talking?"** – be aware of time; step/up step back
- **Avoid pressuring others to speak**
- **Use "I" statements**.
- **Be mindful of privilege and power dynamics**
- **Don't assume we aren't prejudiced**
- **Challenge the idea and not the person.** Respect we're all learning.
- **Understand the difference between intent and impact**
- "I appreciate this feedback."
- "This is very helpful."
- "This is hard, but also important."
- Feeling your heart race? **Take a deep breath or two**.
- **No one knows everything, together we know a lot**

## Appendix B: Interviewee Characteristics based on primary affiliation

| Participant Group | Interviews | Individuals | Roles & Characteristics | Race/Ethnicity | Gender | Location(s) |
|---|---|---|---|---|---|---|
| Farmers | 47 | 48 | Animals farmed: Dairy ($n=13$); Diversified livestock ($n=20$); Beef cattle ($n=9+1$ diversified); Poultry ($n=1+9$ diversified); Pigs ($n=1+4$ diversified); Goats ($n=1+\geq3$ diversified); Sheep ($n=2+\sim3$ diversified) | BIPOC ($n=12$); White ($n=36$) | Male ($n=26$); Female ($n=22$) | Primarily Midwest (WI, MN, IL, IA) |
| Civil Society/Non-Profits | 44 | 45 | Nonprofit and some university participants. Focused primarily on supporting BIPOC farmers ($n=9$) | BIPOC ($n=5$); White ($n=40$) | Male ($n=21$); Female ($n=24$) | Primarily regional/national nonprofits based in Midwest |
| Native Nation | 10 | 10 | Roles in tribal land and agriculture management. Animals: buffalo, chickens, pigs, cattle; historically some renting out of land for dairies/row crops but shifting practices. | Majority Indigenous (7 Tribe members) | Male ($n=8$); Female ($n=2$) | Primarily 4 tribes with some or all their land within Wisconsin |
| Government Employees (USDA/NRCS, State, County) | 17 | 17 | NRCS staff ($n=10$, incl. 4 Tribal Liaisons); State agency staff ($n=6$, incl. Dept. of Natural Resources, Dept. of Ag., 1 BIPOC-focused); County-level staff ($n=1$) Tribal Liaison or BIPOC producer focused ($n=5$) | BIPOC ($n=3$); White ($n=14$) | Male ($n=10$); Female ($n=7$) | Government offices/agencies across Midwest states |
| Other | 8 | 8 | Private sector and academic participants with specialized expertise in agricultural markets, finance, genetics, forage system consulting, business, and dairy economics. | BIPOC ($n=0$); White ($n=8$) | Male ($n=8$); Female ($n=0$) | WI and IL |

**Acknowledgements** This paper emerged from a collective process and reflects the insights, commitments, and learnings shared by more than one hundred people who took part in interviews and workshops across the U.S. Midwest. We are deeply grateful to the farmers, farmworkers, Tribal representatives, nonprofit staff, researchers, educators, and agency professionals who contributed their time, experience, and creativity to this work. Participants represented a range of organizations and communities, including the American Farmland Trust, Practical Farmers of Iowa, Green Lands Blue Waters, the Wisconsin Tribal Conservation Advisory Council, the Oneida Nation, the Ho-Chunk Nation, Forest County Potawatomi, the Illinois Stewardship Alliance, the Wisconsin Farmers Union, the Regenerative Agriculture Alliance, the Land Stewardship Project, Marbleseed, the Michael Fields Agricultural Institute, the University of Wisconsin, and many independent and diversified farms throughout the region. We also thank the facilitation and coordination team for their efforts in organizing the participatory approaches and supporting analysis, transcription, and translation. In particular, we appreciate the contributions of Sarah Lloyd, Andrew Bernhardt, Claudia Calderón, Greta Landis, Laura Paine, Mia Ljung, Michelle Miller, Michael Bell, Randall Jackson, and Rebecca Power. Thank you to Alex Lara, Gaby Shay, and Ryan Hellenbrand for transcription and translation, as well as Liz Anna Kozik for bringing our learnings to life through her illustrations. This work was supported by the USDA Sustainable Agriculture Systems grant 965 (#2019-68012-29852), the University of Wisconsin Graduate Research Fellowship program, and the University of Wisconsin-Madison Center for Culture, History, and Environment, and the Center for Integrated Agricultural Systems. We are also grateful to the anonymous reviewers whose thoughtful feedback strengthened this paper.

**Open Access** This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

## References

Adelle, C. 2019. The role of knowledge in food democracy. *Politics and Governance* 7(4): 214–223.

Adelman, C. 1993. Kurt Lewin and the origins of action research. *Educational Action Research* 1(1): 7–24.

Akom, A. 2011. Black emancipatory action research: integrating a theory of structural racialisation into ethnographic and participatory action research methods. *Ethnography & Education* 6(1): 113–131.

Altieri, M. A. 1999. The ecological role of biodiversity in agroecosystems. *Agriculture, Ecosystems & Environment* 74(1–3): 19–31.

Altieri, M. A., C. I. Nicholls, A. Henao, and M. A Lana. 2015. Agroecology and the design of climate change-resilient farming systems. *Agronomy for Sustainable Development* 35(3): 869–890.

Bain, J., N. Harden, S. Nordrum, and R Olive. 2021. Cultivating Powerful Participation: Reflections from a food justice and facilitation learning experience. *Journal of Agriculture, Food Systems, and Community Development* 11(1): 59–80.

Baptist, E. E. 2014. *The half has never been told: Slavery and the making of American capitalism*. Basic Books.

Bentley, J. W. 1994. Facts, fantasies, and failures of farmer participatory research. *Agriculture and Human Values* 11(2–3): 140–150.

Biernacki, P. and D Waldorf. 1981. Snowball sampling: Problems and techniques of chain referral sampling. *Sociological Methods & Research* 10(2): 141–163.

Bilotta, G., R. Brazier, and P Haygarth. 2007. The impacts of grazing animals on the quality of soils, vegetation and surface waters in intensively managed grasslands. *Advances in Agronomy* 94: 237–280.

Blanchette, A. 2020. *Porkopolis: American animality, standardized Life, and the factory farm*. Duke University Press.

Blattner, C. 2020. just transition for agriculture? A critical step in tackling climate change. *Journal of Agriculture, Food Systems, and Community Development* 9(3): 53–58.

Braun, C. L., V. Bitsch, and A. M Häring. 2022. Behind the scenes of a learning agri-food value chain: lessons from action research. *Agriculture and Human Values* 39(1): 119–134.

Brydon-Miller, M., D. Greenwood, and P Maguire. 2003. Why Action Research? *Action Research (London)* 1(1): 9–28.

Burkholder, J., B. Libra, P. Weyer, S. Heathcote, D. Kolpin, P. S. Thorne, and M Wichman. 2007. Impacts of waste from concentrated animal feeding operations on water quality. *Environmental Health Perspectives* 115(2): 308–312.

Cadieux, K. and R Slocum. 2015. What does it mean to do food justice? *Journal of Political Ecology* 1–26.

Christens, B. D., V. Faust, J. Gaddis, P. T. Inzeo, C. S. Sarmiento, and S. M Sparks. 2016. Action research. In Handbook of methodological approaches to community-based research : Qualitative, quantitative, and mixed methods, eds., L. A. Jason and D. S. Glenwick. Oxford University Press.

Climate Justice Alliance. n.d. *Just transition*. https://climatejusticealliance.org/just-transition

Cuéllar-Padilla, M. and Á Calle-Collado. 2011. Can we find solutions with people? Participatory action research with small organic producers in Andalusia. *Journal of Rural Studies* 27(4): 372–383.

Erickson, B. H. 1979. Some problems of inference from chain data. *Sociological Methodology* 10: 276.

Escobar, A. 1995. *Encountering Development: The making and unmaking of the Third World*. Princeton University Press.

Escobar, A. 2015. Degrowth, postdevelopment, and transitions: a preliminary conversation. *Sustainability Science* 10(3): 451–462.

Fals-Borda, O. and M. A Rahman. 1991. *Action and knowledge: Breaking the monopoly with participatory action research*. Apex Press.

Fochesatto, A. 2025. Who gets to farm? Reshaping just transitions in (agri)culture. In Post-carbon futures. 1st ed., 107–124. Routledge.

Fox, C., R. Fuentes, F. O. Valdez, G. Purser, and K Sexsmith. 2017. *Milked: Immigrant Dairy farmworkers in New York state*. Workers': Center of Central New York.

Franzluebbers, A. J., L. K. Paine, J. R. Winsten, M. Krome, M. A. Sanderson, K. Ogles, and D Thompson. 2012. Well-managed grazing systems: A forgotten hero of conservation. *Journal of Soil and Water Conservation* 67(4): 100A–104A.

Freire, P. 1970. *Pedagogy of the oppressed*. Continuum.

Gadsden, G. I., N. Golden, and N. C Harris. 2023. Place-based bias in environmental scholarship derived from social-ecological landscapes of fear. *Bioscience* 73(1): 23–35.

Garnett, T., C. Godde, A. Muller, E. Röös, P. Smith, J De Boer I, and Van Zanten H. H. E. 2017. *Grazed and confused?: Ruminating on cattle, grazing Systems, methane, nitrous oxide, the soil carbon sequestration question-and what it all means for greenhouse gas emissions*. https://library.wur.nl/WebQuery/wurpubs/529441.

Gilbert, J. 2015. *Planning democracy: Agrarian intellectuals and the intended New Deal*. Yale University Press.

Gillon, S., E. G. Booth, and A. R Rissman. 2016. Shifting drivers and static baselines in environmental governance: challenges for improving and proving water quality outcomes. *Regional Environmental Change* 16(3): 759–775.

Grande, S. 2018. Refusing the university. In Toward what justice? 47–65. Routledge.

Guthman, J. 2008. "If They Only Knew": Color Blindness and Universalism in California Alternative Food Institutions. *The Professional Geographer: The Journal of the Association of American Geographers* 60(3): 387–397.

Guthman, J. 2011. *If they only Knew": The unbearable Whiteness of alternative food*. MIT Press.

Hall, B. 2011. Towards a knowledge democracy movement: Contemporary trends in community-university research partnerships. *Rizoma Freireano* 9: 1.

Hammelman, C., K. Reynolds, and C. Z Levkoe. 2020. Toward a radical food geography praxis: integrating theory, action, and geographic analysis in pursuit of more equitable and sustainable food systems. *Human Geography* 13(3): 211–227.

Hardie Hale, E., C. C. Jadallah, and H. L Ballard. 2022. Collaborative research as boundary work: learning between rice growers and conservation professionals to support habitat conservation on private lands. *Agriculture and Human Values* 39(2): 715–731.

Harrahill, K., Á. Macken-Walsh, and E O'Neill. 2023. Prospects for the bioeconomy in achieving a just transition: perspectives from Irish beef farmers on future pathways. *Journal of Rural Studies* 100: 103020.

Harrison, J. L. and S. E Lloyd. 2012, 03. Illegality at Work: Deportability and the Productive New Era of Immigration Enforcement. *Antipode* 44(2): 365–385.

Hendrickson, M. K., P. H. Howard, E. M. Miller, and D. H Constance. 2020. The food system: Concentration and its impacts. A special report to the family farm action alliance. https://rules.house.gov/sites/democrats.rules.house.gov/files/Concentration%20and%20Options%202020%20Final%209%2015.pdf.

Herzon, I., R. Mazac, M. Erkkola, T. Garnett, H. Hansson, M. Jonell, M. Kaljonen, T. Kortetmäki, M. Lamminen, A. Lonkila, M. Niva, A.-M. Pajari, T. Tribaldos, M. Toivonen, H. L. Tuomisto, K. Koppelmäki, and E Röös. 2024. Both downsizing and improvements to livestock systems are needed to stay within planetary boundaries. *Nature Food* 5(8): 642–645.

Holleman, H. 2017. De-naturalizing ecological disaster: colonialism, racism and the global Dust Bowl of the 1930s. *The Journal of Peasant Studies* 44(1): 234–260.

Holmes, S. M. 2013. *Fresh fruit, broken bodies*. University of California Press.

Holt-Giménez, E. 2015. Racism and Capitalism: Dual Challenges for the Food Movement. *Journal of Agriculture, Food Systems, and Community Development* 5(2): 23–25.

Holt-Giménez, E. and B Harper. 2016. Food-Systems-Racism: From Mistreatment to Transformation. *Food First* 1(1). https://foodfirst.org/wp-content/uploads/2016/03/DR1Final.pdf.

Horrigan, L., R. S. Lawrence, and P Walker. 2002. How sustainable agriculture can address the environmental and human health harms of industrial agriculture. *Environmental Health Perspectives* 110(5): 445–456.

Horst, M. and A Marion. 2019. Racial, ethnic and gender inequities in farmland ownership and farming in the U.S. *Agriculture and Human Values* 36(1): 1–16.

Hubbard, T. 2014. Chapter 13. Buffalo Genocide in Nineteenth-Century North America: "kill, skin, and sell. In Colonial genocide in Indigenous North America. 292–305. Duke University Press.

Huntington, H. P. 2000. Using traditional ecological knowledge in science: Methods and applications. *Ecological Applications: A Publication of the Ecological Society of America* 10(5): 1270–1274.

Ibarra, A., A. Kulwiec, and G Austin-Nichols. 2023. *Dairy workers study: Legal and community Assessment*. University of Wisconsin-Madison. https://schoolforworkers.wisc.edu/wp-content/uploads/sites/795/2023/06/Dairy-Workers-Study-Legal-and-Community-Needs-Assessment.pdf.

Isenberg, A. C. 2000. *The destruction of the bison: An environmental history, 1750-1920*. Cambridge University Press.

Justice, D. H. and J. M O'Brien. 2022. *Allotment stories: Indigenous land relations under settler siege*. U of Minnesota Press.

Justice Migrant. n.d.. *Milk with dignity*. Retrieved from https://migrantjustice.net/milk-with-dignity.

*Just Transition Principles*. n.d.. Just Transition Alliance; The Just Transition Alliance. Retrieved August 17, 2024, from https://jtalliance.org/what-is-just-transition/.

Kaljonen, M., T. Kortetmäki, and T Tribaldos. 2023. Introduction to the special issue on just food system transition: Tackling inequalities for sustainability. *Environmental Innovation and Societal Transitions* 46(100688): 100688.

Kimmerer, R. 2013. Braiding Sweetgrass. Milkweed Editions.

Klassen, S., L. Medland, P. Nicol, and H Pitt. 2023. Pathways for advancing good work in food systems. *Journal of Agriculture, Food Systems, and Community Development* 12(2): 249–265.

Lee, R. and T Ahtone. 2020, March 30. *Land-grab universities*. https://www.hcn.org/issues/52-4/indigenous-affairs-education-land-grab-universities/. High Country News.

Lerner, S. 2010. Sacrifice zones. *The Front Lines of Toxic Chemical Exposure in the United States* 368.

Löhr, K., M. Weinhardt, and S Sieber. 2020. The "World Café" as a participatory method for collecting qualitative data. *International Journal of Qualitative Methods* 19: 160940692091697.

Lowe, Erin B., Ana Fochesatto, and Adena R. Rissman. 2023. Managed Grazing and Agroecological Transformation in the Midwestern United States. *Frontiers in Sustainable Food Systems* 7(August): 1096230.

Matias, C. E. 2016. *Feeling white: Whiteness, emotionality, and education*. 1st ed. [PDF] Sense. https://doi.org/10.1007/978-94-6300-450-3.

Minkoff-Zern, L. A. 2019. *The new American farmer: Immigration, race, and the struggle for sustainability*. MIT Press.

Méndez, V. E., M. Caswell, S. R. Gliessman, and R Cohen. 2017. Integrating agroecology and participatory action research (PAR): Lessons from Central America. *Sustainability (Switzerland)* 9(5): 705–705.

Murphy, S. P., S. M. Cannon, and L Walsh. 2022. Just transition frames: Recognition, representation, and distribution in Irish beef farming. *Journal of Rural Studies* 94: 150–160.

Nash, M. A. 2019. Entangled pasts: Land-grant colleges and American Indian dispossession. *History of Education Quarterly* 59(4): 437–467.

Newell, P. and D Mulvaney. 2013. The political economy of the 'just transition'. *The Geographical Journal* 179(2): 132–140.

Nordling, N. and R Pugh. 2019. Beyond the 'usual suspects' - Alternative qualitative methods for innovation policy studies. *African Journal of Science Technology Innovation and Development* 11(4): 513–522.

Oates, L. G., D. J. Undersander, C. Gratton, M. M. Bell, and R. D Jackson. 2011. Management-intensive rotational grazing enhances forage production and quality of subhumid cool-season pastures. *Crop Science* 51(2): 892–901.

O'Hara, P. 2009. *Enhancing stakeholder participation in national forest programmes tools for practitioners*. Food and Agriculture Organization of the United Nations.

Paine, L. K., D. Undersander, and M. D Casler. 1999. Pasture growth, production, and quality under rotational and continuous grazing management. *Journal of Production Agriculture* 12(4): 569–577.

Peeters, A. 2015. Participatory and holistic approaches with grassland farmers and development of policies. *D. Vijay et Al* 290–305.

Peeters, A. 2019. Participatory and holistic approaches with grassland farmers and development of policies. *International Grassland Congress Proceedings*. https://uknowledge.uky.edu/igc/23/keynote/33.

Penniman, L. 2018. *Farming while black: Soul fire Farm's practical guide to liberation on the land*. Chelsea Green Publishing.

Rasmussen, L. V., B. Coolsaet, A. Martin, O. Mertz, U. Pascual, E. Corbera, N. Dawson, J. A. Fisher, P. Franks, and C. M Ryan. 2018. Social-ecological outcomes of agricultural intensification. *Nature Sustainability* 1(6): 275–282.

Reynolds, K., and N. Cohen. 2016. *Beyond the kale: Urban agriculture and social justice activism in New York City* (Vol. 28). University of Georgia Press.

Rios, S. M. 2022. Movement as a methodology: Finding the right tools for doing equity-oriented research. *Latino Studies* 20(3): 390–407.

Rissman, A. R., A. Fochesatto, E. B. Lowe, Y. Lu, R. M. Hirsch, and R. D Jackson. 2023. Grassland and managed grazing policy review. *Frontiers in Sustainable Food Systems* 7. https://doi.org/10.3389/fsufs.2023.1010441.

Saldaña, J. 2021. *The coding manual for Qualitative researchers*. 4th ed. SAGE Publications.

Slocum, R. 2006. Anti-racist Practice and the Work of Community Food Organizations. *Antipode* 38(2): 327–349.

Smith, L. T. 1999. *Decolonizing methodologies: Research and indigenous peoples*. London ; New York; Dunedin; New York: Zed Books : University of Otago Press : Distributed in the USA exclusively by St Martin's Press.

Sovacool, B. K. 2021. Who are the victims of low-carbon transitions? Towards a political ecology of climate change mitigation. *Energy Research & Social Science* 73: 101916.

Spivak, G. C. 1988. Can the subaltern speak? In Marxism and the interpretation of culture, eds., C. Nelson and L. Grossberg, 271–313. Urbana: University of Illinois Press.

Spratt, E., J. Jordan, J. Winsten, P. Huff, C. van Schaik, J. G. Jewett, M. Filbert, J. Luhman, E. Meier, and L Paine. 2021. Accelerating regenerative grazing to tackle farm, environmental, and societal challenges in the upper Midwest. *Journal of Soil and Water Conservation* 76(1): 15A–23A.

Stark, A., F. Gale, and H Murphy-Gregory. 2023. Just transitions' meanings: A Systematic Review. *Society & Natural Resources* 36(10): 1277–1297.

Stoecker, R. 2009. Are we talking the walk of community-based research? *Action Research (London)* 7(4): 385–404.

Stoecker, R. and A Falcón. 2022. *Handbook on participatory action research and community development*. Edward Elgar Publishing.

Swilling, M. 2020. *The age of Sustainability; just transitions in a complex World*. Routledge.

Swords, A. 2019. Action research on organizational change with the Food Bank of the Southern Tier: a regional food bank's efforts to move beyond charity. *Agriculture and Human Values* 36(4): 849–865.

Tsing, A. L. 2015. *The mushroom at the end of the World: On the possibility of Life in capitalist ruins*. Princeton University Press.

U.S. Department of Agriculture, National Agricultural Statistics Service. 2022. *Dairy farms in Wisconsin*. USDA NASS Quick Stats. Retrieved from https://quickstats.nass.usda.gov.

Wang, X. and K Lo. 2021. Just transition: A conceptual review. *Energy Research & Social Science* 82: 102291.

Welton, S. and J Eisen. 2019. Clean energy justice: Charting an emerging agenda. *The Harvard Environmental Law Review: HELR* 43(2): 307–371.

Wepking, C., H. C. Mackin, Z. Raff, D. Shrestha, A. Orfanou, E. G. Booth, C. J. Kucharik, C. Gratton, and R. D Jackson. 2022. Perennial grassland agriculture restores critical ecosystem functions in the U.S. Upper Midwest. *Frontiers in Sustainable Food Systems* 6: 1010280.

Wezel, A., S. Bellon, T. Doré, C. Francis, D. Vallod, and C David. 2009. Agroecology as a science, a movement and a practice. A review. *Agronomy for Sustainable Development* 29(4): 503–515.

Williams, S. and A Doyon. 2019. Justice in energy transitions. *Environmental Innovation and Societal Transitions* 31: 144–153.

Wittman, H., A. Desmarais, and N. Wiebe, Eds. 2010. *Food sovereignty: Reconnecting food, nature and community*. Fernwood Publishing.

Yates, L. 2015. Rethinking prefiguration: Alternatives, micropolitics and goals in social movements. *Social Movement Studies* 14(1): 1–21.

**Publisher's Note** Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

**Ana Fochesatto** is a postdoctoral researcher in Community and Environmental Sociology at the University of Wisconsin-Madison. Her research interests lie at the intersection of food, justice, and social change with an emphasis on community-based research for just transitions in agriculture.

**Erin B. Lowe** is a project coordinator for the University of Wisconsin-Madison and the Great Lakes Intertribal Food Coalition, supporting Tribally-led efforts to build Indigenous food sovereignty. Erin earned her MS and PhD in Agroecology and Sustainable Food Systems at UW-Madison, conducting research on pollinator conservation, crop pollination, and socially just managed grazing systems in the Midwest.

**Adena R. Rissman** is a professor in the Department of Forest and Wildlife Ecology at the University of Wisconsin-Madison, where she leads the People, Institutions, and Ecosystems (PIE) Lab. Her interdisciplinary research examines the human and policy dimensions of ecosystem, natural resources, and agricultural management.